

# v1.0Vibrational Transference Deep Dive

## V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

**Architecture:** One-Vibration Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026  
**Applies to:** v1.0Vibrational Transference Deep Dive V2.0.md

### Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines’ Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables  $10^{7+}$  particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

### V1.0 vs V2.0 Specific Changes

| Parameter                  | V1.0                        | V2.0                                                                                                                 | Severity | Consequence if Not Updated                                                          |
|----------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------|
| Room-T quantum coherence   | Quantum biology speculative | Renger 2023 Nature Chemistry: quantum coherence in FMO complex at room-T confirmed. COHERENT mode enabled $T < 310K$ | HIGH     | Room-temperature quantum coherence validated — no longer speculative for simulation |
| Phonon quantum engineering | Conceptual bath             | IBM Quantum 2022: phonon bath engineering for qubit protection. PHONON_BATH object with tunable $T_{eff}$            | MEDIUM   | Engineered quantum resonance becomes precise simulation tool                        |
| DNA soliton measurement    | Yakushevich predicted       | 2022 attosecond spectroscopy: DNA kink soliton at $v_s \sim 10^3$ m/s measured. DNA-SOLITON particle type added      | HIGH     | Measured biological soliton — adds biological domain to vibrational simulation      |

| Parameter                             | V1.0                    | V2.0                                                                                                          | Severity | Consequence if Not Updated                                             |
|---------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------|
| <b>Kuramoto brain synchronization</b> | Order parameter concept | MEG 2024: gamma synchronization $r \sim 0.7$ during memory encoding. Visual: phase-locked particles glow gold | MEDIUM   | Neurological measurement grounds Kuramoto implementation in experiment |

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## Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** DNA, Room-T
  - **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
  - **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
  - **New interaction modes:** Triggered by new experimental discoveries
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## Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time  $\tau$  as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

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*End of Versioning Delta | v1.0Vibrational Transference Deep Dive V1.0 archived | V2.0 is the active specification as of June 2026*